Indian Institute of Information Technology, Pune

B.Tech Minor Project II Presentation

Skin Cancer Detection Using Deep Transfer Learning

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Problem Statement

- **Skin cancer** is one of the most common forms of cancer worldwide.
- **Early detection** of skin cancer is crucial for effective treatment and improved survival rates.
- **Dermatologists** typically diagnose skin cancer through **visual inspection** of **skin lesions** using **dermoscopic images**.
- **Manual diagnosis** has several limitations:
  - It can be **subjective** due to varying human interpretations.
  - It is often **time-consuming**.
  - It is **prone to human error**, especially when distinguishing between **benign** (non-cancerous) and **malignant** (cancerous) lesions that appear visually similar.

Basal cell carcinoma: This is the most common type of skin cancer. It looks like a flesh-colored, pearl-like bump, or pinkish patch of skin.

Squamous cell carcinoma: The second most common type of skin cancer. Often looks like a red firm bump, scaly patch, or a sore that heals and then re-opens.

Actinic keratoses: These dry, scaly patches or spots are precancerous growths.

Melanoma: The deadliest form of skin cancer. Frequently develops in a mole or suddenly appears as a new dark spot on the skin.

Proposed Solution

- With the rise of **Artificial Intelligence (AI)** and **Deep Learning (DL)**, **automated skin cancer detection systems** have become a key research focus.
- **Deep learning models**, especially **Convolutional Neural Networks (CNNs)**, have shown **outstanding performance** in image classification tasks.
- CNNs are capable of **automatically learning complex visual features** from input images without the need for manual feature extraction.
- However, **training deep neural networks from scratch** poses several challenges:
  - It requires **large labeled datasets**, which are often unavailable in medical imaging.
  - It demands **high computational resources**.
- To address these challenges, this project utilizes **Deep Transfer Learning**.
- In transfer learning:
- **Pre-trained models** such as **EfficientNet B3**, **ResNet-50**, or **DenseNet-169** originally trained on large-scale datasets like **ImageNet**—are **fine-tuned** on **dermoscopic images** from the **ISIC dataset**.
- This approach allows the model to **reuse learned visual features** from general images and **adapt them** to the specific task of **skin lesion detection**.

Clinical Impact: Automated systems can reduce diagnostic errors and enable earlier intervention, directly improving patient outcomes and treatment success rates.

Motivation

Skin cancer is among the **most prevalent cancers globally**, with cases increasing significantly over the past few decades due to prolonged exposure to ultraviolet (UV) radiation, lifestyle changes, and environmental factors.

- According to the **World Health Organization (WHO)**, **1 in every 3 cancers diagnosed is a skin cancer**.
- Over **2–3 million non-melanoma** and **132,000 melanoma** skin cancer cases are reported annually worldwide.
- **Melanoma**, although less common (about **1% of all skin cancers**), is the **most deadly**, accounting for the majority of skin cancer deaths.
- In **India**, skin cancer cases are also on the rise, with an estimated **30,000+ new cases annually**, though underreporting remains a concern due to lack of awareness and screening infrastructure.
- The **five-year survival rate** for early-detected melanoma is above **95%**, but it drops drastically to around **25%** for late-stage detection — showing the critical need for early and accurate diagnosis.

These statistics highlight the **urgency for automated diagnostic systems** that can assist dermatologists in detecting skin cancer at an early stage, especially in regions with limited medical expertise or resources.

Real World Applications

- **Clinical Decision Support Systems (CDSS):** AI models can assist dermatologists by providing second opinions, highlighting suspicious regions, and suggesting possible diagnoses from dermoscopic images.

- **Mobile Health (mHealth) Applications:** Smartphone-based apps integrated with deep learning models enable users to capture images of skin lesions and get instant risk assessments, promoting self-screening and early awareness.

- **Teledermatology:** In remote or rural areas where access to specialists is limited, AI-powered telemedicine platforms allow patients to upload images for remote evaluation by dermatologists assisted by AI models.

Literature Review

Author(s) & Year	Objective	Methodology (Models, Datasets)	Key Findings & Performance	Identified Limitations / Relevance to Project
Ahmed et al. (2024)	To enhance skin cancer classification accuracy by fusing multiple deep learning models using attention mechanisms.	Models: Multi-Model Ensemble (ResNet50V2, MobileNetV2, EfficientNetV2). Dataset: HAM10000, ISIC.	The "Attentional Fusion" method achieved superior accuracy compared to single models by dynamically focusing on the most relevant features of the lesion.	Directly validates our project's approach of using an Ensemble of ResNet, DenseNet, and EfficientNet to improve prediction reliability over single models.
Goyal et al. (2020)	To automate the removal of common artifacts (hair, gel bubbles) and background skin through precise segmentation to improve classification results.	Models: Ensemble of Mask R-CNN and DeepLabV3+. Dataset: ISIC 2017, ISIC 2018.	Automated segmentation significantly reduced false positives caused by background noise. The ensemble method outperformed individual segmentation models in boundary detection.	Highlights the critical importance of our Preprocessing Module (artifact removal and segmentation) before feeding images into the classifier to ensure data quality.
Hasan et al. (2021)	To perform a comparative analysis of different CNN architectures for benign vs. malignant classification.	Models: MobileNetV2, Xception, ResNet50. Dataset: HAM10000.	While MobileNetV2 offered efficiency, deeper networks like ResNet provided better feature extraction for complex, similar-looking lesions.	Confirms that for high-stakes medical diagnosis, deeper architectures like ResNet-50 are preferred over lighter models for feature extraction capabilities.
Hoang et al. (2023)	To address the severe class imbalance problem in skin cancer datasets to improve the detection of the minority class (malignant cases).	Method: Proposed "Categorical Weighting Domination" (a weighted loss function approach). Dataset: ISIC Archive.	The method achieved significantly higher Sensitivity and Recall for malignant cases compared to standard Cross-Entropy loss.	Strongly supports our decision to use Focal Loss to handle the inherent imbalance between benign and malignant images in our dataset.

Author(s) & Year	Objective	Methodology (Models, Datasets)	Key Findings & Performance	Identified Limitations / Relevance to Project
Kassem et al. (2020)	To classify skin lesions into eight distinct classes using deep transfer learning.	Models: GoogleNet (Inception) with Transfer Learning. Dataset: ISIC 2019.	Achieved 94.92% accuracy, proving that models pre-trained on ImageNet can effectively learn specific dermatological features.	Demonstrates the robustness of CNNs on ISIC datasets and supports the scalability of our project to multi-class classification in the future.
Kausar et al. (2021)	To classify multi-class skin lesions using transfer learning and fine-tuning on pre-trained models.	Models: Ensemble of ResNet, InceptionV3, and DenseNet. o Dataset: HAM10000.	Demonstrated that fine-tuning pre-trained weights on specific dermatological data yields significantly better generalization than training models from scratch.	Justifies our choice of DenseNet and ResNet as base learners and confirms the absolute necessity of Transfer Learning for this domain.
Nigar et al. (2022)	To solve the "Black Box" problem in medical AI by implementing Explainable AI (XAI) techniques for skin lesion classification.	Models: Xception, ResNet50. o XAI Tools: LIME and Grad-CAM. o Dataset: HAM10000.	The model achieved high accuracy while providing visual heatmaps that accurately highlighted the lesion borders, thereby increasing clinical trust in the model's decisions.	Justifying the integration of Grad-CAM to make our system transparent and trustworthy for doctors.
Wei et al. (2020)	To develop a lightweight detection system suitable for mobile or edge devices with limited computational resources.	Models: Ensemble Lightweight Network (based on CNN architectures). o Dataset: Dermoscopy Image Database.	The proposed lightweight model achieved >99% accuracy while maintaining significantly lower computational complexity compared to heavier networks.	Validates the industry need for efficient architectures. This supports our project's use of EfficientNet, which is specifically designed to balance parameter diagno

Research Gap 1 : Class Imbalance in Skin Cancer Detection

Issue: Prevalent Class Imbalance

- **Medical imaging datasets are often imbalanced** — benign lesions greatly outnumber malignant ones. Models may achieve high accuracy but fail to detect malignant (minority) cases, significantly reducing their clinical usefulness and potentially leading to missed diagnoses.

Solution 1: Data-Level Techniques

- **Data Augmentation:** Use techniques like rotation, flipping, and zooming to create more samples for the minority class, increasing dataset diversity.
- **SMOTE / Random Oversampling:** Systematically generate synthetic samples or duplicate existing minority samples to balance the class distribution.

Solution 2: Algorithm-Level Techniques

- **Class-Weighted Loss:** Assign higher penalties to misclassifications of the minority class during training, forcing the model to pay more attention to these crucial examples.
- **Focal Loss:** Modifies the standard cross-entropy loss to down-weight easy examples and focus learning on hard, misclassified malignant cases, preventing dominance by abundant benign cases.

Research Gap 2 : Skin Tone Variability & Metadata

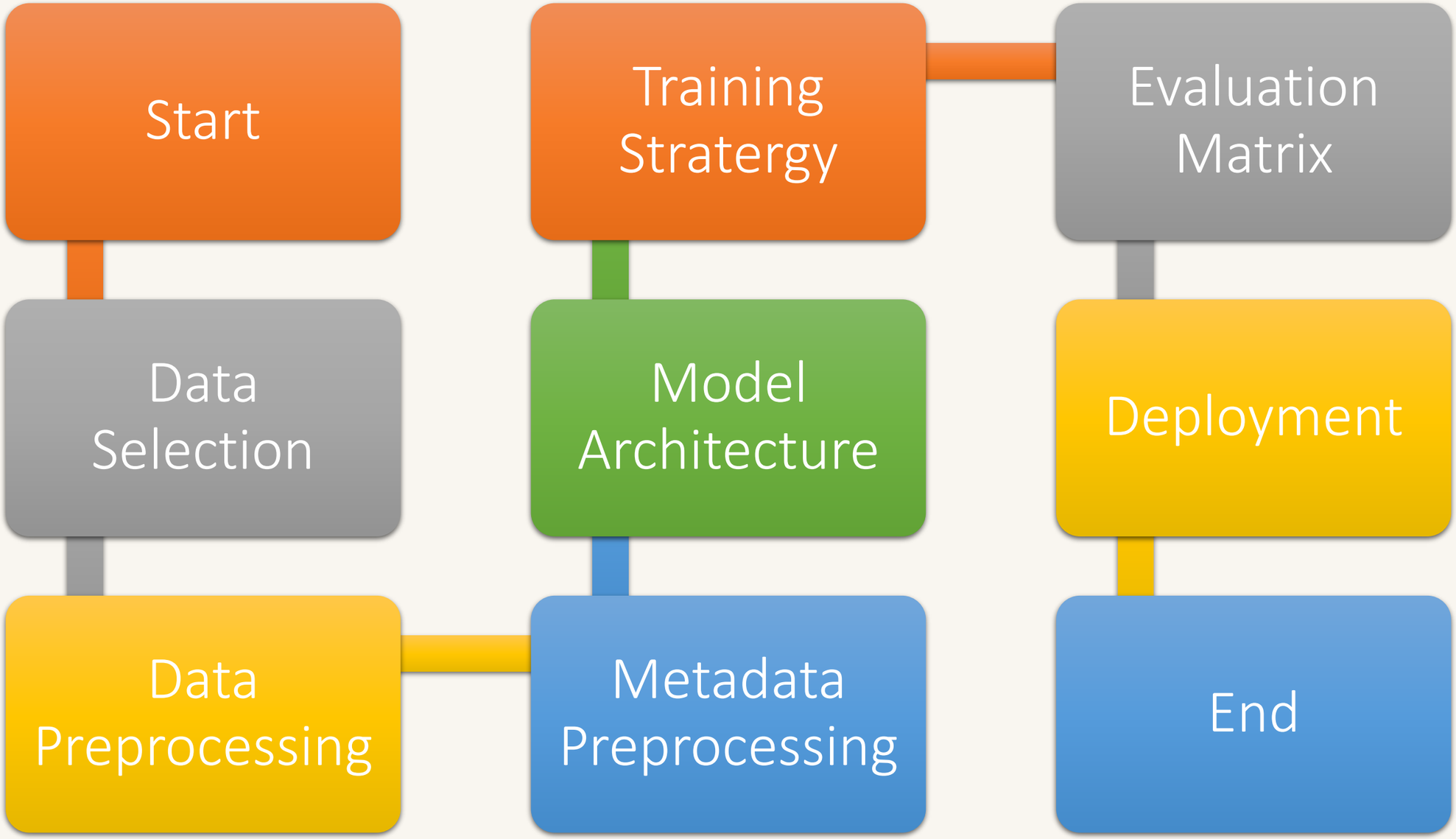
Skin Tone Variability

- Models often trained on lighter skin tones, leading to poor generalization to darker tones.
- This results in biased predictions and reduced diagnostic accuracy across diverse populations.
- Solution: Use diverse, balanced datasets representing all skin types.

Metadata Availability

- Lack of patient metadata (age, gender, lesion location, etc.) limits contextual learning.
- Metadata can improve diagnostic accuracy when combined with image data.
- Solution: Encourage standardized datasets with integrated metadata.

Methodology for Skin Cancer Detection



DATASET SELECTION

Name:

1. ISIC 2019 , ISIC 2020
2 . FANCONIC dataset

Size:

1. Total Benign : ~50,000
2. Total Malignant : ~5000

Description

The dataset comprises dermatologist-annotated Dermoscopic images of skin lesions, classified as either benign or malignant, and is primarily used for skin cancer detection and diagnostic classification tasks.

Metadata:

Age, Anatomical Site , Sex , Lesion_ID

Data Preprocessing

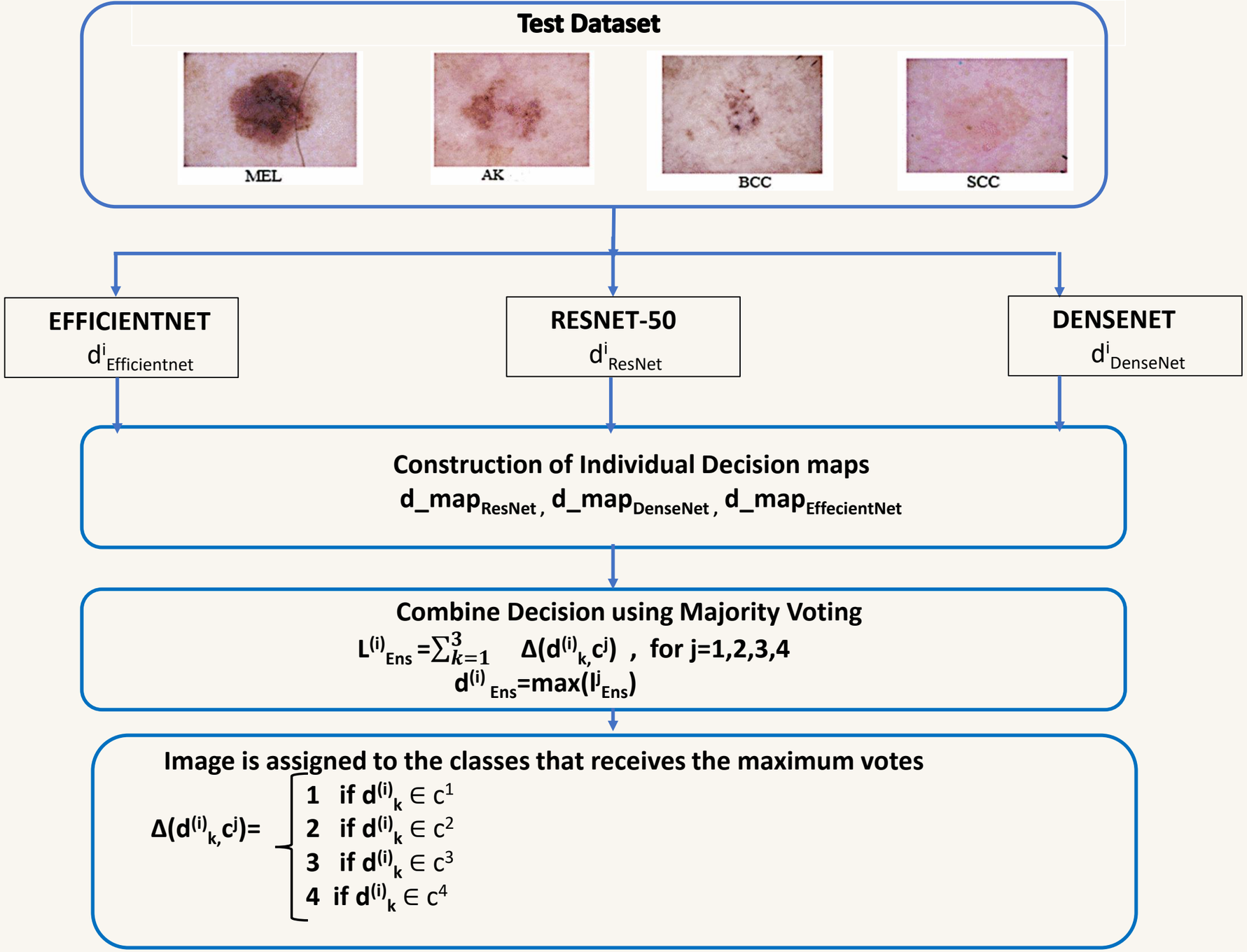
- Standardize and augment Dermoscopic images.
- Resize all images to 224×224 pixels.
- Applied image augmentations: horizontal/vertical flips, random rotations, color jitter, and zooms.

Metadata Preprocessing

- Fill missing ages using median imputation
- Bin or normalize age values
- One-hot encode categorical variables (sex, lesion location)
- Min-max scaling
- Standard scaling (zero mean, unit variance)

Model Architecture

- **Base Models:** EfficientNet-B3, DenseNet-169, ResNet-50
- **Transfer Learning:** Fine-tune models on dataset to adapt to skin features.
- **Classification Head:**
  - Input layer
  - Convolutional layer
  - Activation function
  - Pooling layer
  - Fully-connected layer


Test Dataset
MEL
AK
BCC
SCC
EFFICIENTNET
$d^i_{Efficientnet}$
RESNET-50
d^i_{ResNet}
DENSENET
$d^i_{DenseNet}$
Construction of Individual Decision maps
d_map_{ResNet} , $d_map_{DenseNet}$, $d_map_{EffecientNet}$
Combine Decision using Majority Voting
$L^{(i)}_{Ens} = \sum_{k=1}^{3} \Delta(d^{(i)}_k, c^j)$, for j=1,2,3,4
$d^{(i)}_{Ens} = \max(l^j_{Ens})$
Image is assigned to the classes that receives the maximum votes
$\Delta(d^{(i)}_k, c^j) = \begin{cases} 1 & \text{if } d^{(i)}_k \in c^1 \\ 2 & \text{if } d^{(i)}_k \in c^2 \\ 3 & \text{if } d^{(i)}_k \in c^3 \\ 4 & \text{if } d^{(i)}_k \in c^4 \end{cases}$

How the Models Actually Work

ResNet-50, EfficientNet-B3, and DenseNet are Convolutional Neural Networks (CNNs) that process images through the following steps:

1. Input Image

The model takes an image as input (like a dermoscopic photo).

2. Feature Extraction with Convolutions

Multiple filter layers scan the image to detect basic features like edges, colors, shapes, and textures.

3. Layer Stacking and Learning Patterns

Through many layers, the model learns complex patterns by combining simpler features - edges form shapes, shapes identify lesion characteristics.

4. Special Connections and Ensembling the model

- **ResNet-50:** Uses residual connections that skip layers to prevent information loss during training.
- **DenseNet:** Connects each layer to every other layer for better information flow and feature reuse.
- **EfficientNet B!/B4:** Carefully balances network size and depth for optimal learning with efficient computing.

5. Prediction

After processing through all layers, the model outputs a classification - benign or malignant lesion.

Training Strategy

Dataset Split
Train, validation and test sets

Transfer Learning:
Used pretrained model

Fine-Tuning
Unfroze top layers for fine-tuning with low learning rate

Loss Function :
Categorical cross-entropy

Optimizer :
Adam with learning rate scheduling

Regularization :
Dropout & early stopping to reduce overfitting

Metrics Tracked :
Accuracy, precision, recall

Evaluation Metrics

Accuracy

Percentage of correctly classified samples over total samples

Precision

Proportion of predicted positive samples that are truly positive

Recall

Proportion of actual positives correctly identified

F1 Score

Harmonic mean of precision and recall

Confusion Matrix

Table of true positives, false positives, true negatives, false negatives.

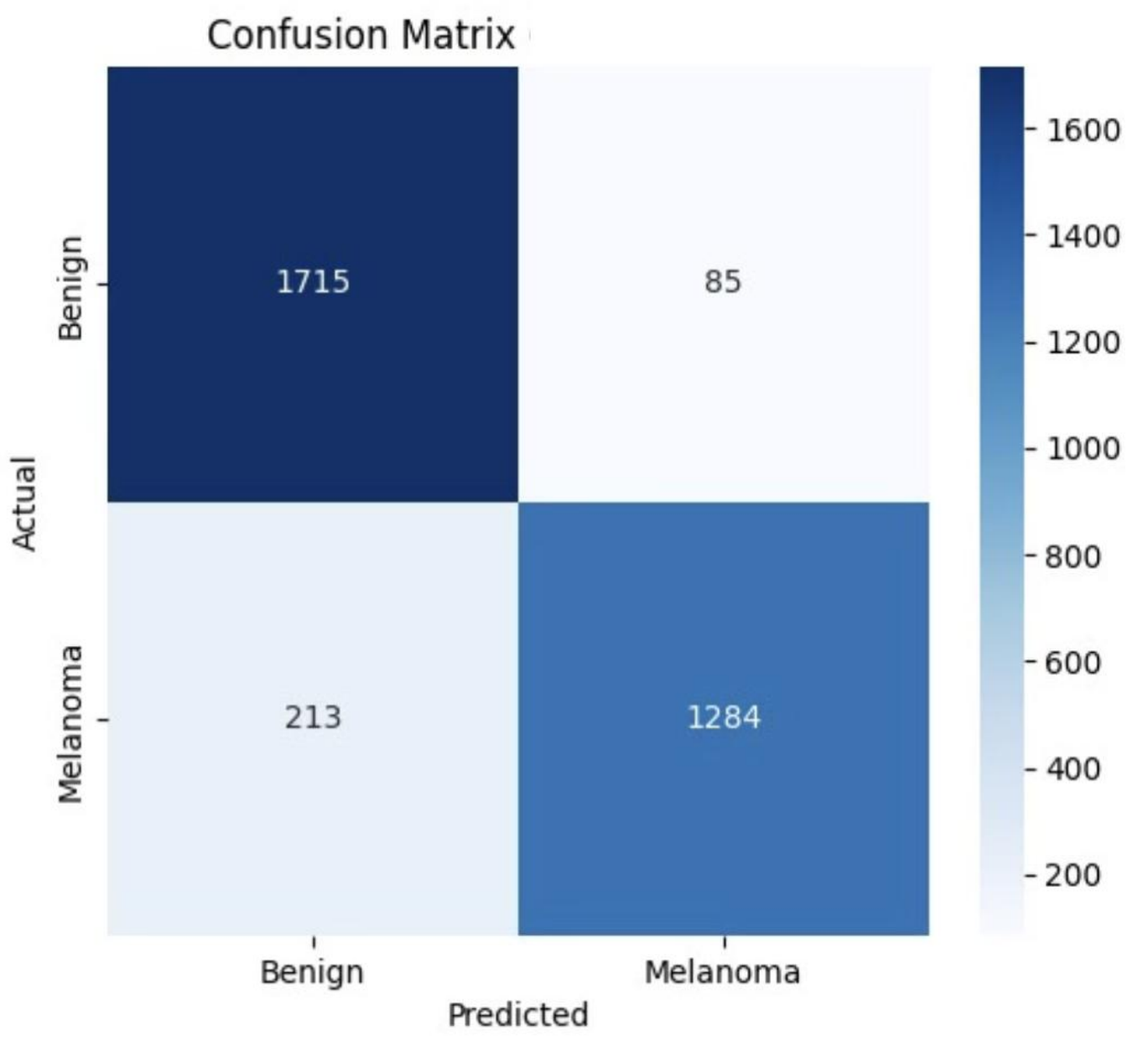
ROC Curve & AUC

Measures trade-off between true positive rate and false positive rate.

Results

Model	F1-Score	Accuracy	Precision	Recall(Sensitivity)	Balanced Accuracy
DenseNet 121	78.1%	83.5%	97.6%	65.2%	81.95%
EfficientNet B1	61.8%	74.7%	98.4%	45.1%	72.2%
EfficientNet B4	81.5%	85.6%	97.5%	70.0%	84.3%
ResNet 50	75.2%	81.9%	96.7%	62.3%	80.3%

Results



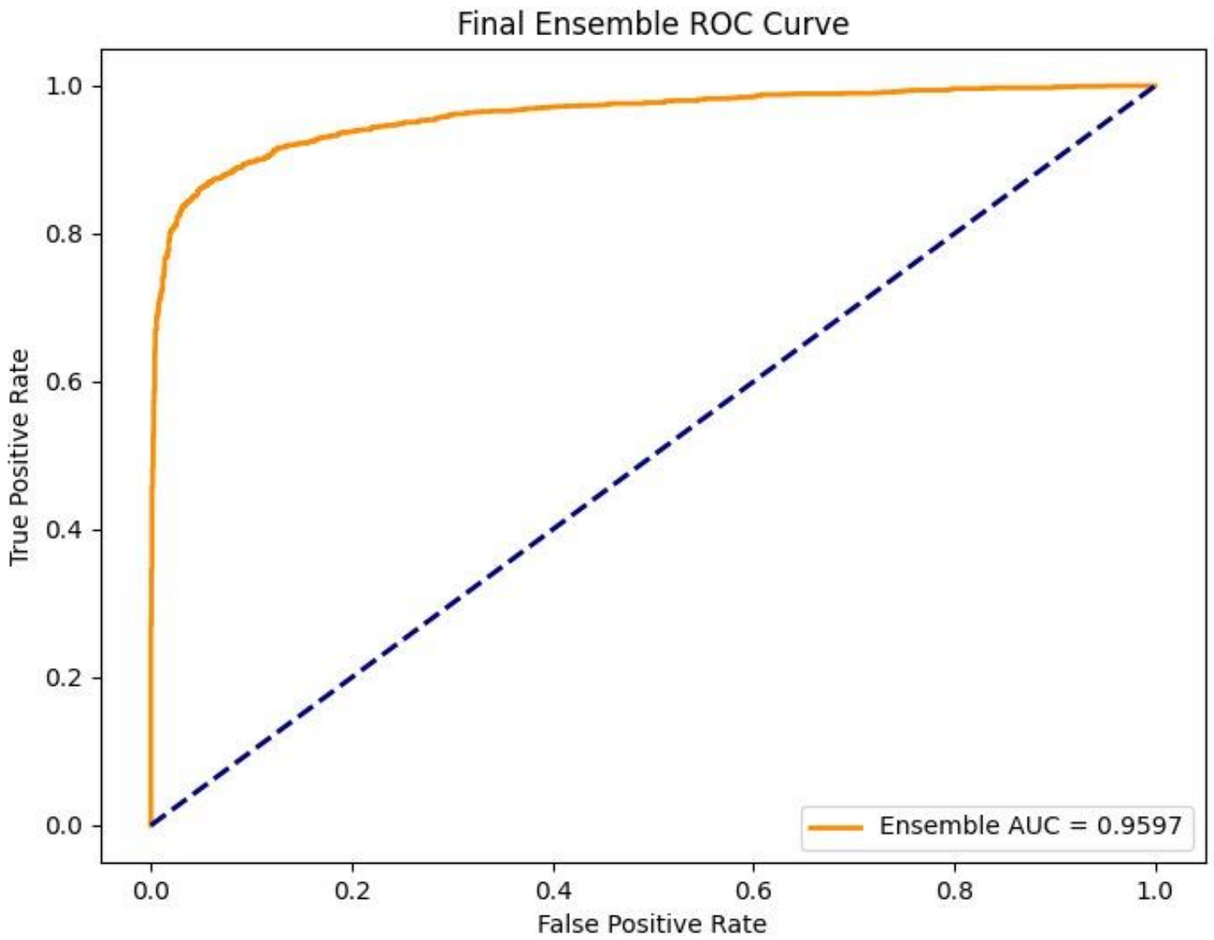


Fig: Ensemble Model Confusion Matrix

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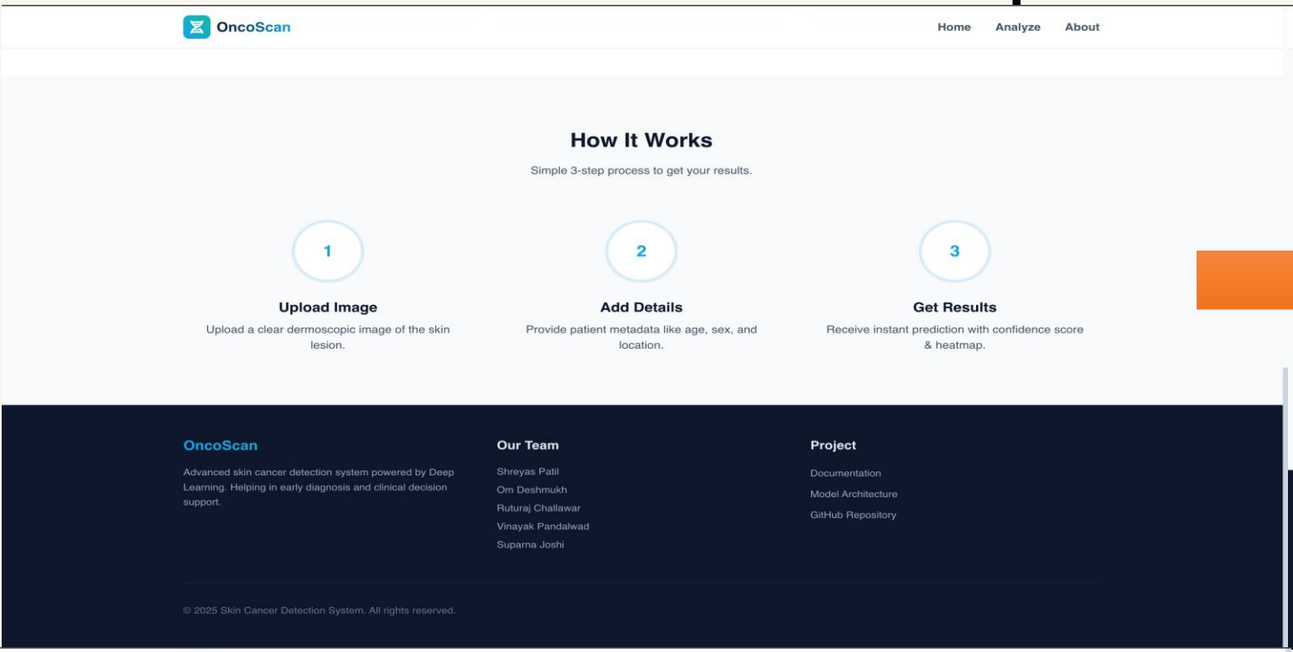
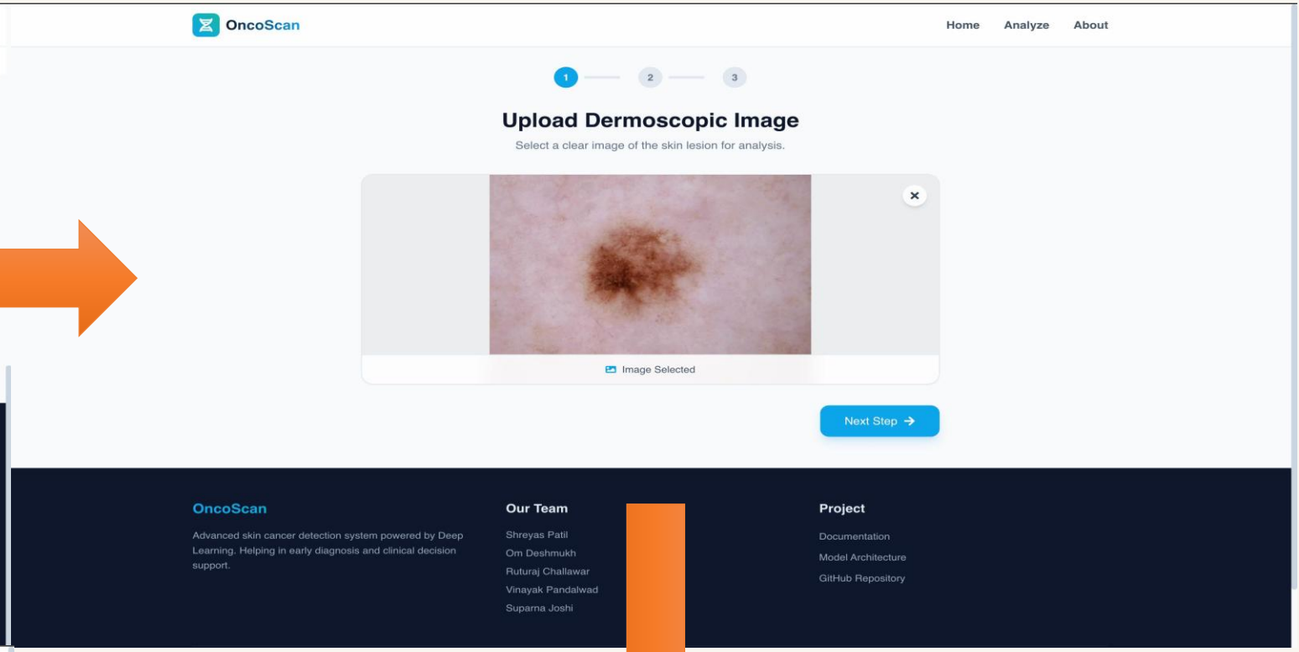
Results

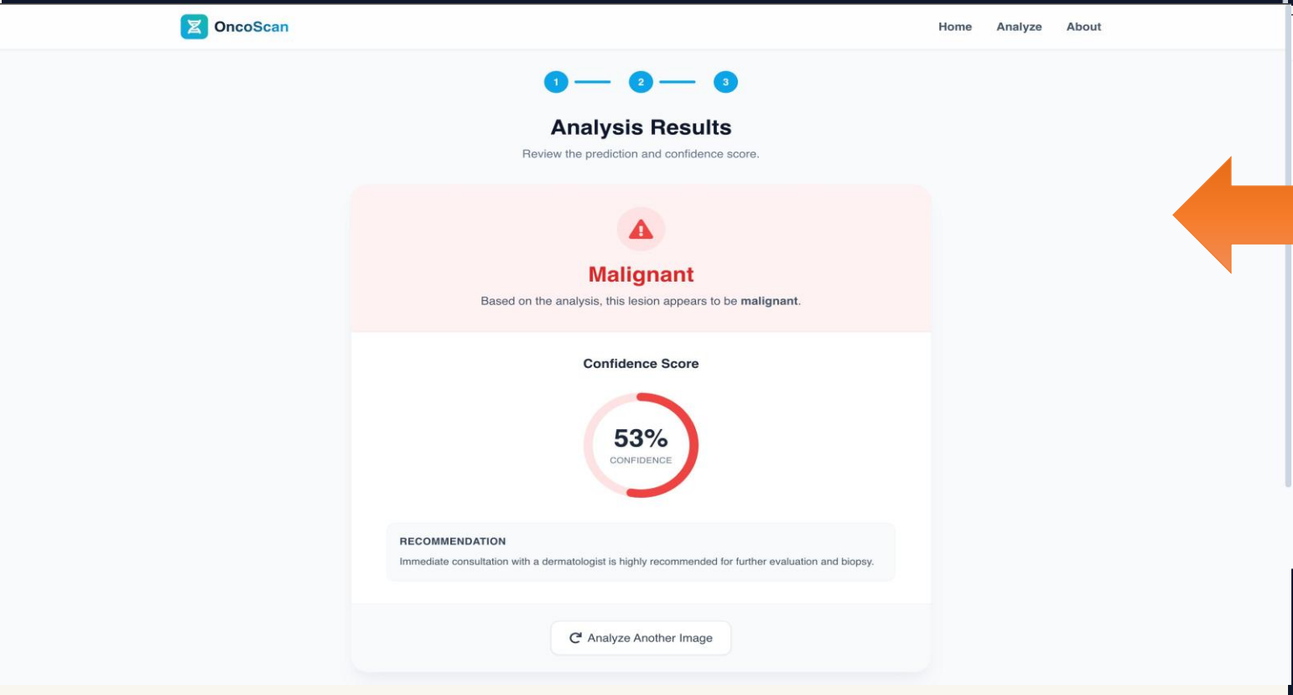
	F1 Score	Accuracy	Precision	Recall
Ensemble Model	**89.60%**	**90.96%**	**93.78%**	**85.76%**

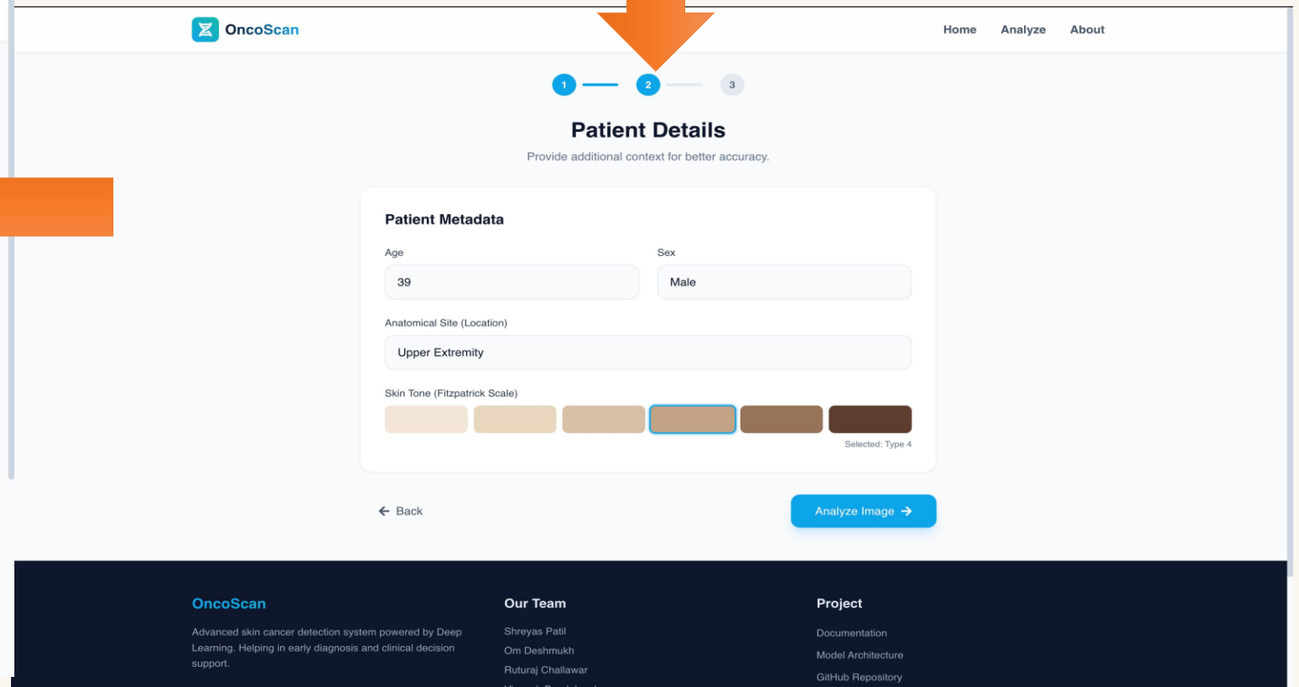
Optimal threshold :0.28
Explanation: If the predicted value of the image is greater than 0.28 then the image is classified as malignant **.**

Developed Solution









Future Scope

Grad-CAM

Purpose

Grad-CAM (Gradient-weighted Class Activation Mapping) is used in Explainable AI (XAI) to visualize where a deep learning model focuses when diagnosing skin cancer.

How It Works

- Computes gradients of the target class (e.g., malignant) with respect to the final convolutional layer
- Generates a heatmap highlighting the most important regions influencing the prediction
- Red areas → strong model attention; blue areas → weak attention

Benefits

- Improves transparency and trust in AI predictions
- Helps clinicians validate model decisions
- Aids in identifying model bias or errors in lesion interpretation

Future Scope

Multi-Class Classification

- Expand beyond binary classification (benign vs. malignant) to detect multiple lesion types such as melanoma, basal cell carcinoma (BCC), squamous cell carcinoma (SCC), and actinic keratosis (AK).
- This will help the system identify and tell apart different types of skin cancers more accurately, similar to how a doctor diagnoses various conditions.
- Requires larger, balanced, and well-annotated datasets.

Attention Mechanisms

- Integrate attention modules (e.g., CBAM, SE-Block, or Vision Transformers) into CNN architectures.
- Helps the model focus on the most important parts of the lesion instead of unnecessary background areas.
- Expected to increase accuracy, explainability, and trust in model predictions.

Conclusion

- The primary objective of this project was to develop an automated, robust, and clinically interpretable system for the early detection of skin cancer using Deep Learning. Through this study, we have successfully implemented an ensemble deep learning framework that addresses the critical limitations of manual diagnosis.
- The proposed **ensemble deep-learning model** for melanoma detection demonstrated strong and reliable performance across all evaluation metrics. The final model achieved an overall **accuracy of 90.96%**, indicating a high level of correctness in classifying skin lesions as benign or melanoma. The most **optimal threshold is 0.28** which is used to classify the image into benign or malignant.
- By integrating four powerful Convolutional Neural Networks : **EfficientNet-B1, EfficientNet-B4, ResNet-50 and densenet121** we leveraged the power of Transfer Learning to extract complex features from images. A key contribution of this work was the implementation of **Focal Loss** and **Activation Function**, which effectively mitigated the issue of class imbalance inherent in the **ISIC 2019 and 2020** datasets.
- In conclusion, this project establishes that a deep learning-based **Clinical Decision Support System (CDSS)** can serve as a powerful second opinion for dermatologists.

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